

Physics 340 Final Exam: Mathematical Physics

Instructions

Answer all five questions. Show all work for full credit. Each question is worth 20 points.

Question 1: Algebra of Vectors (20 points)

Consider the following two vectors in $\mathbb{R}^3$:

$$\mathbf{A} = 2\hat{i} - \hat{j} + 4\hat{k}, \quad \mathbf{B} = -\hat{i} + 3\hat{j} + 2\hat{k}.$$

- a) Compute the dot product $\mathbf{A} \cdot \mathbf{B}$.
- b) Compute the cross product $\mathbf{A} \times \mathbf{B}$.
- c) Find the angle between the two vectors $\mathbf{A}$ and $\mathbf{B}$.

Continue your solution here

Question 2: Coupled Linear First-Order Differential Equations (20 points)

Consider the coupled system:

$$\frac{dx}{dt} = 4x + y, \quad \frac{dy}{dt} = 2x + 3y.$$

a) Write the system in matrix form:

$$\frac{d\vec{X}}{dt} = A\vec{X}.$$

b) Find the eigenvalues and eigenvectors of the matrix A .

c) Solve the system for $x(t)$ and $y(t)$, given

$$x(0) = 1, \quad y(0) = 2.$$

d) Describe, in words, the long-term behavior of the system.

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Question 3: AC RC Circuit and Complex Impedance (20 points)

An alternating voltage source

$$V(t) = V_0 e^{i\omega t}$$

is applied across a resistor R and a capacitor C connected in series.

- a) Using complex impedance, derive the total impedance Z of the RC circuit in terms of R , C , and ω .
- b) Using the complex form of Ohm's law,

$$V(t) = I(t)Z,$$

find an expression for the current $I(t)$.

- c) Find the phase shift between the voltage and current. State whether the current leads or lags the voltage.
- d) Briefly explain the physical role of the resistor and the capacitor in producing this phase shift.

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Question 4: Fourier Analysis (20 points)

Let $f(x)$ be periodic with period $2L$, defined on one period by

$$f(x) = \begin{cases} -1, & -L < x < 0, \\ 1, & 0 < x < L. \end{cases}$$

- a) Compute the Fourier coefficients a_0 , a_n , and b_n .
- b) Write the Fourier series representation of $f(x)$.
- c) Discuss the convergence of the series at:
 - i) points where $f(x)$ is continuous,
 - ii) points where the periodic extension is discontinuous.

Question 5: Vector Calculus (20 points)

Let

$$\mathbf{F} = (xy)\hat{i} + (x^2 + z)\hat{j} + (y + z^2)\hat{k}.$$

a) Compute the divergence:

$$\nabla \cdot \mathbf{F}.$$

b) Compute the curl:

$$\nabla \times \mathbf{F}.$$

c) Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

along the curve

$$\mathbf{r}(t) = \langle t, t^2, t \rangle, \quad 0 \leq t \leq 1.$$

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